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Lecture -21
LR as Prediction

- $F = \{ f(x) = \beta_0 + \beta_1 x + \ldots + \beta_p x_p \}$

$L(y, a) = (y-a)^2$

high bias, low variance, escapes the curse of dimensionality

- $y \in R^n$, $X \in R^{n \times (p+1)}$

$\min. \; RSS(\beta) = \|y - X\beta\|^2 = (y-X\beta)^T (y-X\beta)$

$\frac{\partial RSS}{\partial \beta} = 0 \Rightarrow X^T X\hat{\beta} = X^T y$

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T y}$$

$\hat{y} = X\hat{\beta}$

$= \underbrace{X (X^T X)^{-1} X^T}_{H} y$

- Least sq. as project^n

$H \rightarrow$ orthogonal projec^n on col. space of X.

$H = H^T$ $\quad H^2 = H$, $\quad X^T (y - \hat{y}) = 0$

$\hat{y}$ doesn't to y

$\hat{y} - y$ is 0 (residual is orthogonal)

$tr(H) = p+1$

- if X does not have a full column rank, when $p > n$, then $X^T X$ is singular
- Simple regression $\rightarrow \hat{\beta}_1$ describes the relation of x & y, ignoring everything.

Multiple $\rightarrow \hat{\beta}_1$ describes the relation of x & y holding the other predictors fixed. (partial effect).

- When cols. of X are nearly depend., $X^T X$ is singular

then $\hat{\beta}$ has var., coefficients s with small data changes & indi. coeff. are meaningless.

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but $\hat{y}$ is stable, predic^n at new X in same reg. is fine,

- least sq. $\hat{y} = Hy$ $\quad H = X(X^TX)^{-1}(X^T)$

KNN

The fitted value at x_i is

$$\hat{y}_i = \frac{1}{K} \sum_{l \in N_K(x_i)} y_l$$

$\hat{y} = Sy$ with $S_{il} = \begin{cases} 1/K & x_l \text{ is a neighbor of } x_i \\ 0 & \text{o.w.} \end{cases}$

both are linear smoothers.

- least sq. is a orthogonal projec^n onto the col. space of X.
- coefficient is a partial effect, it flips signs relative to its marginal version. There is no such thing as the effect of a var., only its effect given what else is in the model.
- Collinearity destabilizes coefficients while leaving predictions intact. Thus model is good for predic^n but useless for interpreta^n.